

ESSEX COUNTY AP, NJ, USA

WMO: 724094

Lat: 40.876N

Lon: 74.283W

Elev: 53

StdP: 100.69

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.3	-22.2	0.5	-10.9	-19.7	0.7	-8.1	9.5	-1.6	8.8	-2.4	2.8	320	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.5	23.2	32.1	22.5	30.6	21.9	24.9	30.2	24.1	28.8	23.5	27.9	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.4	26.9	22.7	17.6	26.2	22.2	17.0	25.7	75.7	30.3	72.7	28.8	69.9	27.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	5.8	-15.5	36.0	2.8	1.6	-17.5	37.2	-19.1	38.2	-20.7	39.1	-22.7	40.3		
(5)			-16.4	26.2	2.6	1.1	-18.2	27.0	-19.7	27.6	-21.2	28.2	-23.1	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-1.1	0.2	4.6	11.0	16.5	21.4	24.2	23.3	19.4	12.8	7.1	2.0	(6)
(7)		DBStd	9.83	5.51	4.99	5.00	4.82	4.32	3.58	2.99	2.95	3.68	4.47	4.59	4.78	(7)
(8)		HDD10.0	1228	346	275	178	42	2	0	0	0	22	110	253		(8)
(9)		HDD18.3	2916	603	507	424	226	88	12	1	2	31	178	337	507	(9)
(10)		CDD10.0	1907	1	2	13	72	204	341	442	412	282	111	23	4	(10)
(11)		CDD18.3	553	0	0	0	6	33	104	184	156	63	8	0	0	(11)
(12)		CDH23.3	4979	0	1	3	97	344	947	1757	1319	464	47	1	0	(12)
(13)		CDH26.7	1756	0	0	0	29	112	326	698	463	123	6	0	0	(13)
(14)	Wind	WSAvg	2.1	2.5	2.6	2.8	2.7	2.1	1.9	1.7	1.5	1.6	1.9	2.1	2.4	(14)
(15)	Precipitation	PrecAvg	1209	89	73	104	97	94	110	110	116	101	98	93	102	(15)
(16)		PrecMax	1678	187	157	263	252	229	249	209	349	211	346	164	186	(16)
(17)		PrecMin	863	42	20	17	41	18	27	37	44	25	0	31	27	(17)
(18)		PrecStd	214	36	33	53	47	41	55	44	66	56	78	36	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	17.9	22.8	30.3	32.6	34.2	36.1	35.1	32.7	27.4	22.1	18.6	(19)
(20)			MCWB	13.7	12.8	14.6	18.2	20.8	22.8	23.7	24.5	23.1	20.0	16.8	14.5	(20)
(21)		2%	DB	12.8	13.7	18.8	26.1	29.8	32.2	33.8	32.6	29.9	24.3	19.1	14.5	(21)
(22)			MCWB	10.0	9.1	12.0	16.0	19.9	22.6	23.4	23.2	21.4	18.7	14.9	12.1	(22)
(23)		5%	DB	8.9	10.7	15.6	22.7	27.3	30.1	32.3	31.0	27.7	22.4	17.1	11.9	(23)
(24)			MCWB	6.0	6.3	9.6	14.5	18.7	21.6	22.8	22.3	20.5	17.7	12.7	9.2	(24)
(25)		10%	DB	6.6	7.5	12.6	19.6	24.6	28.4	30.6	29.1	26.1	20.5	14.8	9.3	(25)
(26)			MCWB	3.9	4.1	7.9	12.3	17.4	20.3	22.2	21.4	20.0	16.2	11.1	6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	14.9	15.8	20.0	22.8	24.6	26.2	26.3	24.7	22.6	18.2	15.8	(27)
(28)			MDCB	16.0	17.5	21.0	26.2	29.3	30.6	33.0	32.4	29.5	24.6	20.7	17.3	(28)
(29)		2%	WB	10.5	10.2	13.3	17.7	21.4	23.8	25.1	24.8	23.5	20.7	15.8	12.6	(29)
(30)			MDCB	12.4	13.2	17.5	23.6	27.3	29.5	30.8	29.5	26.9	23.3	18.2	14.1	(30)
(31)		5%	WB	6.7	6.7	10.7	15.8	20.3	22.9	24.3	23.9	22.6	18.8	13.9	9.9	(31)
(32)			MDCB	8.4	9.3	14.2	21.4	25.2	28.2	29.1	28.1	25.7	21.5	16.1	11.4	(32)
(33)		10%	WB	4.1	4.4	8.3	13.8	18.9	21.9	23.5	23.1	21.5	17.0	11.8	6.9	(33)
(34)			MDCB	6.4	7.5	12.2	17.9	23.2	26.5	28.2	26.9	24.6	19.3	13.9	8.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	9.3	9.8	11.0	11.1	10.5	10.4	10.1	10.4	10.3	9.7	7.9	(35)
(36)			MCDBR	11.5	13.6	14.5	16.0	14.4	12.4	12.1	11.6	12.2	12.6	12.4	11.0	(36)
(37)			MCWBR	9.6	10.0	9.0	7.9	6.4	5.3	4.3	4.2	5.5	7.0	8.4	9.5	(37)
(38)		5% WB	MCDBR	10.9	12.4	12.6	13.6	12.3	10.7	9.8	9.7	9.4	9.8	10.1	10.1	(38)
(39)			MCWBR	9.7	10.0	9.3	8.1	6.3	5.4	4.2	4.3	5.3	6.8	8.5	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.324	0.357	0.379	0.425	0.487	0.488	0.474	0.397	0.359	0.327	0.309	(40)	
(41)		taud	2.448	2.406	2.387	2.363	2.262	2.126	2.136	2.167	2.354	2.470	2.518	2.515	(41)	
(42)		Ebn at Noon	849	887	892	893	855	800	794	794	839	837	823	822	(42)	
(43)		Edh at Noon	78	94	108	118	134	154	151	143	111	88	73	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.84	2.74	3.68	4.73	5.25	5.75	5.90	5.11	4.27	2.95	2.07	1.55	(44)	
(45)		RadStd	0.15	0.28	0.30	0.47	0.55	0.53	0.30	0.33	0.44	0.31	0.19	0.15	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+112
(47)	Regional (63 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+67

Nomenclature: See separate page